

Assignment 3: Spatial Prediction and Validation

GG501 – Spatial Knowledge Mobilization
Department of Geography and Environmental Studies
Wilfrid Laurier University

Instructor: J. Goetz

Student Name: Gazi Shaila Afroz

Student ID: 255842150

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1 Introduction

This assignment compares two methods for predicting landslide susceptibility: a generalized additive model (GAM) and a random forest (RF). The goal is to examine how model choice and validation strategy affect predictive performance, especially the difference between cross-validation and spatial cross-validation, and to present the workflow, results, and interpretation of the model comparison.

2 Methods

2.1 Model Comparison

Two models are fitted and compared:

- Generalized Additive Model (GAM)

A Generalized Additive Model (GAM) is a statistical model that allows flexible, nonlinear relationships between predictors and the response variable. It uses smooth functions to represent how environmental variables such as slope, elevation, and rainfall affect landslide occurrence. GAM is useful for landslide susceptibility prediction because it can capture nonlinear effects of terrain variables and is relatively easy to interpret.

- Random Forest (RF)

A Random Forest (RF) is a machine learning method that builds many decision trees and combines their predictions. Each tree is trained using a different subset of the data. The final prediction is based on the average of all trees. Random Forest is useful for landslide susceptibility prediction because it can capture complex relationships between environmental variables and often provides high predictive accuracy.

2.2 Validation Strategy

Model performance is evaluated using:

- Cross-validation (CV)

Cross-validation randomly divides the data into training and testing sets. However, in spatial data nearby observations are often similar because of spatial autocorrelation. This can lead to overly optimistic estimates of model performance.

- Spatial cross-validation (SPCV)

Spatial cross-validation separates the data based on geographic location so that training and testing samples come from different areas. This reduces the effect of spatial autocorrelation and provides a more realistic estimate of how well the model can predict landslides in new locations.

3 Results

3.1 Summary of Predictor Characteristics

Landslide locations tend to occur at lower elevations than non-landslide locations, while also being, on average, closer to roads and deforested areas, suggesting that both topographic position and human disturbance may be associated with landslide occurrence (Table 1). Landslide samples also have slightly higher slopes and larger upslope contributing areas, whereas curvature variables show only small differences between the two groups, indicating weaker contrasts for local surface shape relative to elevation and distance-based predictors.

Table 1: Summary statistics for predictor variables.

Variable	Mean (Landslide)	SD (Landslide)	Mean (Non-landslide)	SD (Non-landslide)
elev	2064.259	250.248	2272.883	314.298
slope	33.747	11.482	32.780	11.098
profcurv	-0.002	0.020	-0.000	0.018
plancurv	-0.003	0.018	0.000	0.017
logcarea	3.235	0.645	3.027	0.623
distroad	193.124	121.355	277.590	64.810
distdeforest	182.227	120.072	250.528	94.477

3.2 Random Forest Variable Importance

The random forest importance measures indicate that distance to roads, elevation, and log contributing area are the most influential predictors, with distance to roads having the highest MeanDecreaseAccuracy and elevation having the highest MeanDecreaseGini (Table 2; Figure 1). This suggests that landslide susceptibility in the study area is driven primarily by a combination of topographic position, hydrological accumulation, and proximity to human disturbance, while slope and the curvature variables appear to play weaker roles relative to these predictors.

Table 2: Random Forest variable importance.

Variable	FALSE	TRUE	MeanDecreaseAccuracy	MeanDecreaseGini
elev	-3.803	59.187	64.353	254.236
slope	5.283	21.004	18.593	107.116
profcurv	6.845	11.270	13.528	106.743
plancurv	1.186	16.142	14.135	108.245
logcarea	16.928	45.107	46.392	140.998
distroad	4.504	77.841	85.071	122.764
distdeforest	-26.400	49.402	45.052	80.888

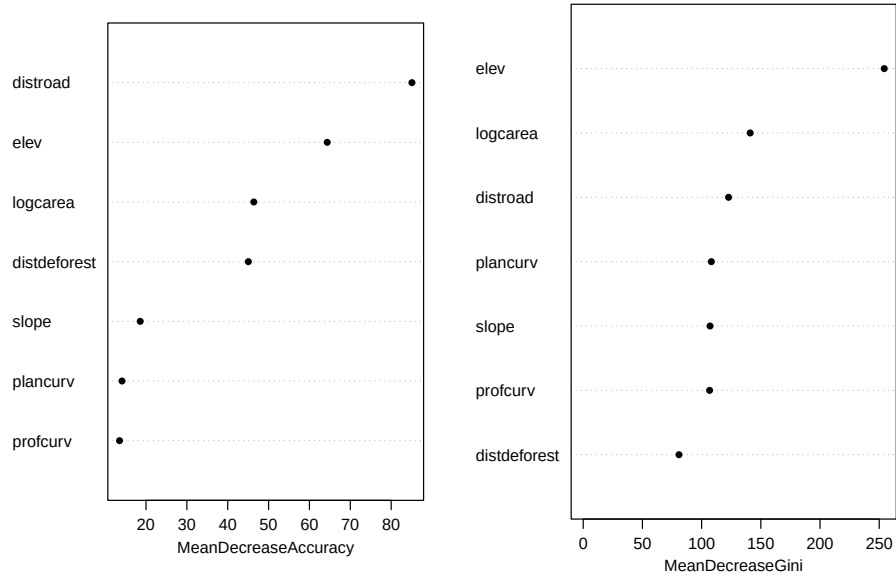


Figure 1: Variable importance from the random forest model.

3.3 GAM Fitted Functions

The GAM smooths indicate that elevation has the strongest nonlinear effect, with landslide susceptibility increasing slightly at mid elevations before declining sharply at higher elevations, while log contributing area shows a curved relationship with higher susceptibility at intermediate values (Figure 2). Distance to roads and distance to deforested areas both show decreasing effects with increasing distance, suggesting landslides are more likely closer

to roads and deforested areas, whereas slope and the curvature variables exhibit comparatively weak or near-linear effects.

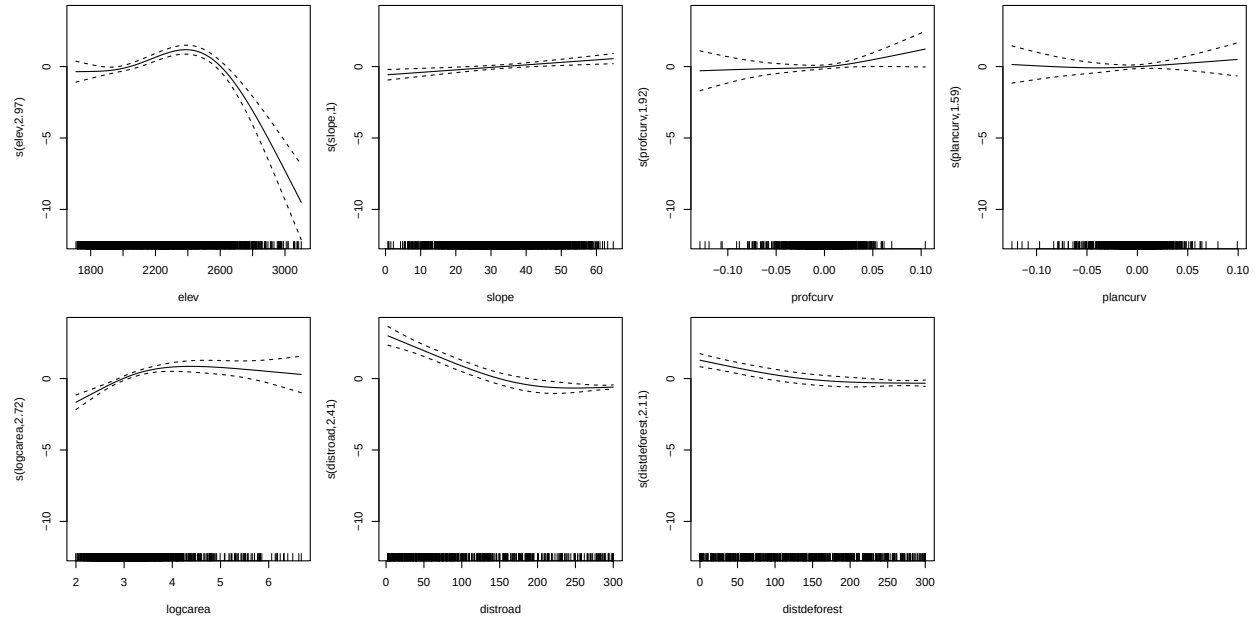


Figure 2: Fitted smoothing functions from the GAM

3.4 Model Performance Comparison

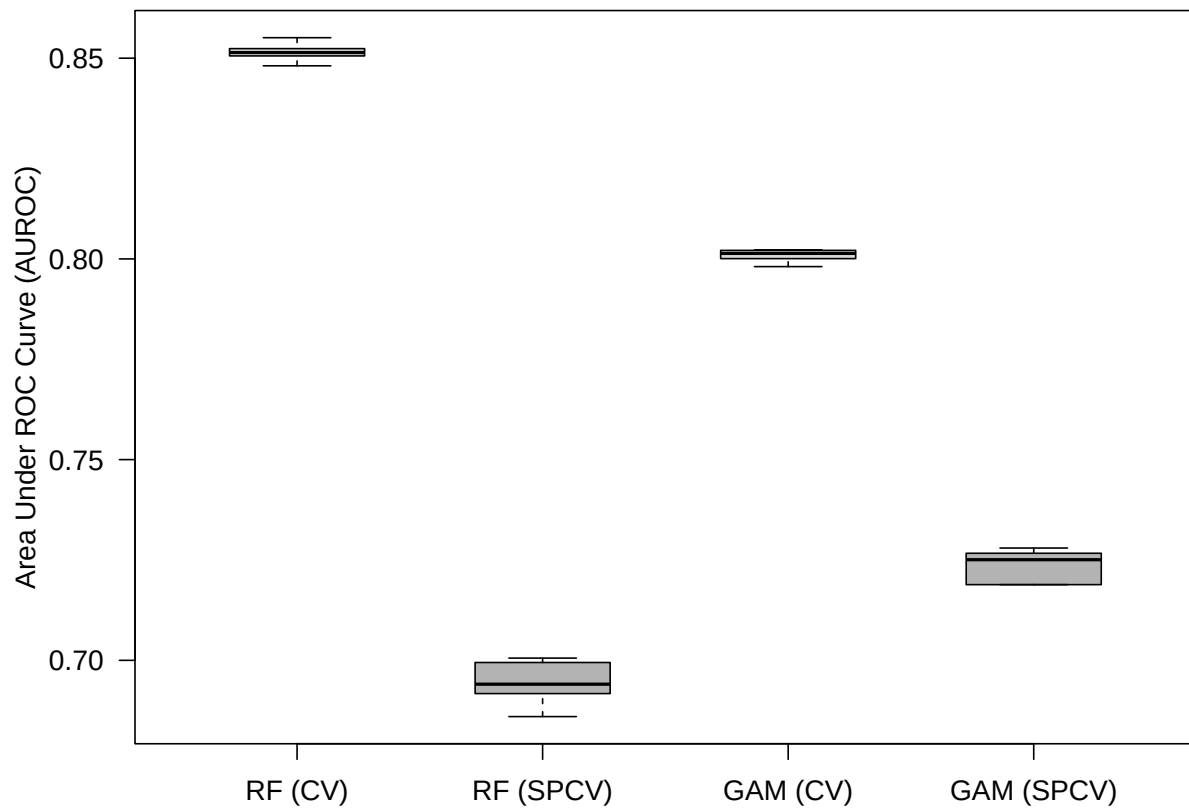


Figure 3: Comparison of GAM and random forest performance under cross-validation and spatial cross-validation.

As illustrated in Figure 3, the random forest performs better than the GAM under cross-validation. However, under spatial cross-validation the GAM shows slightly higher performance. Both models show lower performance with spatial cross-validation than with cross-validation.

3.5 Cross-Validation versus Spatial Cross-Validation Sampling

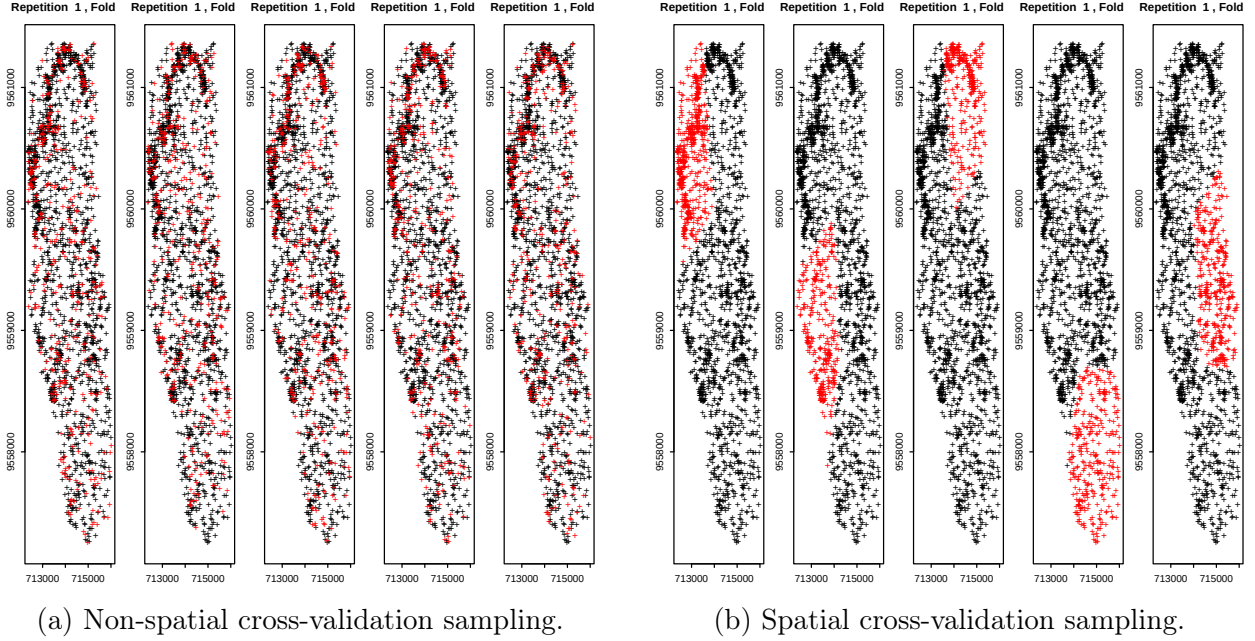


Figure 4: Illustration of non-spatial cross-validation and spatial cross-validation sampling.

As illustrated in Figure 4, in non-spatial cross-validation the observations are randomly divided into training and testing folds across the study area. In contrast, spatial cross-validation divides the observations based on their geographic location so that training and testing data are spatially separated.

3.6 Prediction Maps

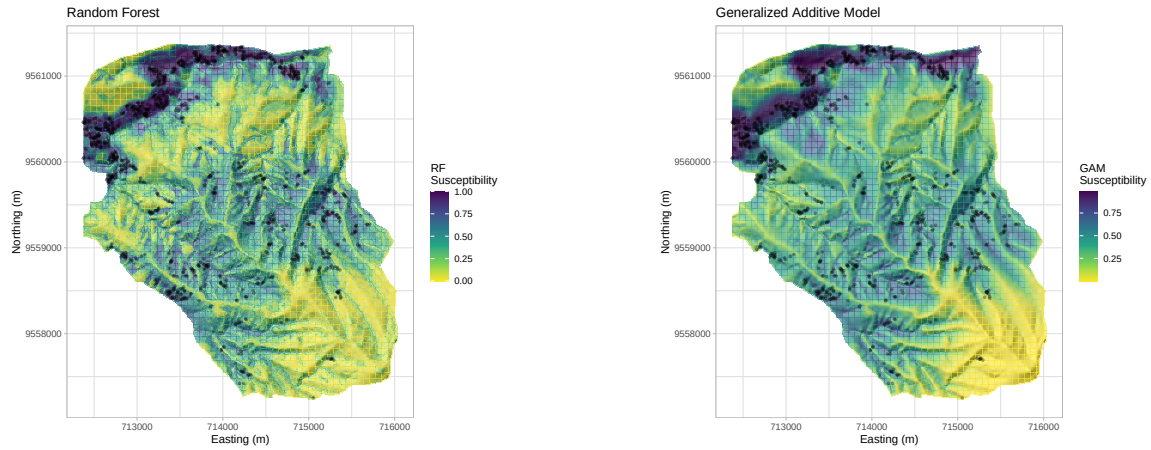


Figure 5: Landslide susceptibility prediction surfaces for the GAM and random forest models, shown with hillshade background for terrain context.

As shown in Figure 5, both the random forest and GAM models predict similar areas of high landslide susceptibility. However, the random forest model shows more detailed spatial variation, while the GAM produces smoother patterns.

4 Discussion

The random forest performed better than the GAM under cross-validation, suggesting stronger predictive ability when samples are randomly distributed. However, when spatial cross-validation was used, performance decreased for both models and the GAM performed slightly better. This suggests that cross-validation may overestimate accuracy because of spatial autocorrelation. The reduction in performance under spatial cross-validation indicates that cross-validation may overestimate model accuracy for spatial prediction problems. Spatial cross-validation therefore provides a more realistic estimate of model transferability to new geographic areas.

One limitation of this analysis is that only a few environmental variables were used, mainly terrain and distance to human disturbances. Including other factors such as geology, soil, or rainfall could improve prediction accuracy. The results also depend on the quality of the landslide data and the spatial resolution of the input data.

5 Conclusion

This assignment compared a generalized additive model (GAM) and a random forest (RF) for predicting landslides. The random forest performed better under cross-validation, while the GAM performed slightly better under spatial cross-validation. Both models showed lower performance under spatial cross-validation, showing that cross-validation can overestimate accuracy when spatial autocorrelation exists. These results show that spatial validation is important because it provides a more realistic estimate of model performance and how well the model can predict landslides in new locations.